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Control, Treatment and Prevention of Carp Fish Infestation with *Chilodonella* Cyprinid in the Conditions of Samarkand Region

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Abstract: This article presents the results of a comprehensive study on the occurrence, epizootological characteristics, diagnosis, treatment, and prevention of chilodonellosis caused by *Chilodonella cyprini* in carp fish (*Cyprinus carpio*) cultivated in artificial water bodies of the Samarkand region, Republic of Uzbekistan. The research focuses on the identification of the main causative agents, including *Chilodonella cyprini*, *Chilodonella piscicola*, and *Chilodonella hexasticha*, their biological characteristics, and their influence on the health condition of fish populations. The study analyzes the major factors contributing to the development and spread of infection, including unfavorable environmental conditions, changes in water temperature, deterioration of hydrochemical parameters, organic pollution of ponds, high stocking density, and weakened physiological state of fish during winter periods. Clinical symptoms and pathological changes associated with the disease were investigated through external examination and microscopic analysis of samples collected from the skin, fins, and gills of infected fish. The effectiveness of therapeutic measures using sodium chloride and malachite green solutions was evaluated, demonstrating their role in controlling parasitic infestation. The obtained results confirm that early diagnosis, appropriate treatment, and systematic preventive measures are



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essential for reducing fish mortality, improving fish health, and ensuring sustainable development of carp aquaculture in the Samarkand region and similar aquatic farming environments.

Keywords: *Chilodonella cyprinid*, *Ch. Piscicola*, *Ch. hexasticha*, *carp fish*, *damage*, *epizootology*, *diagnosis*, *treatment*, *pathogenesis*.

Introduction

In our country, great attention is paid to the development of the fishing sector, among all industries, especially in the Samarkand region, as an important factor in the rapid development and transformation of fisheries into one of the most profitable sectors, improving the standard of living of the population, and providing the domestic market with sustainable food such as fish meat. In accordance with the decree of the president of the Republic of Uzbekistan “on measures to support the fishing sector and improve its efficiency” dated August 29, 2020, PQ-4816 was signed in order to support the fishing sector in the Republic, increase the efficiency of the activities of fishing and fishing farms, ensure rational and effective use of land and water resources in this area and Certain studies have been carried out to study the species of carp-like fish common in our republic, their distribution, and their biological characteristics. Of these, the prevention of the spread of *Chilodonella cyprinid* disease in carp (carp) farms in the Samarkand region, its epizootology, and treatment are urgent issues.

The purpose and objectives of the study. Taking into account the above problems, we aimed at measures to prevent the infestation of carp fish with *Chilodonella cyprinid* (chylodonellosis), epizootology, pathogynesis, symptoms, pathological-anatomical changes, diagnosis, treatment, and treatment of carp fish being fed in artificial bodies of water in the regions of Samarkand.

Research materials and methods. In our studies (of the order Hypostomatida), we found that it is a disease of many species of Basin fish that cause cyprinid infusions in the family Chilodonnellidae, of the genus *Chilodonella*, which parasitize on jabra, skin, reptile and other parts of the body. Invasion is manifested by damage to the skin and gill apparatus of fish, and the causative agents of *Chilodonella cyprinid* are divided into two types, the first of which is *Ch. Piscicola*, the second *Ch.Hexasticha* the fact that the first of them is common and has a larger size the body is leafy in shape, flattened in the dorsoventral direction, the back has a cut at the dilated end clinical (swelling of the victim of a gray stain on the body of a fish), microscopic (a sample from the mucous membrane and victim on the skin of The dimensions of the parasite's body - (30-97)×(26-72) mkm-we found in our examination that there are lashes in the form of oblique rows, the number of rows is different, 7-15 on the right side of the body, and 8-14 on the left side.

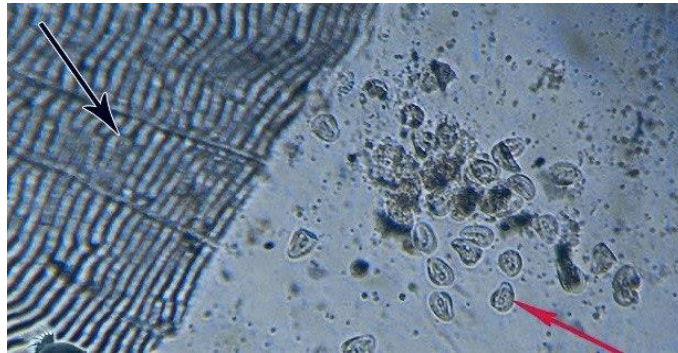
The results obtained and their analysis. Once the hildonellas enter the fish's body, they settle on the skin. They feed on mucus and detached epithelial cells. We found that the parasite's cilia move along the fish's body. We observed that the cyst formation process occurs by division over a period of 3.5-4 hours, with the most rapid multiplication occurring at water temperatures of 5-10°C. Under unfavorable conditions, infusoria can survive for a long time in water or mud. The disease occurs mainly in the winter months of 3-4 months, as well as in ponds of fish farms, fish factories, and spawning and rearing farms. Epizootics of chilodonellosis are not recorded in natural



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reservoirs, but river fish, almost all species of freshwater fish, are carriers of the causative agent of the disease. Chilodonellosis occurs in fish farms with high fish density and causes significant



economic damage to farms due to fish mortality during the winter months. The disease is most often observed in one-year-old fish during the winter months. Fish of other ages can be carriers of the parasite. In summer, the disease does not manifest itself and the chilodonellae disappear, and at this time they turn into cysts in a dormant state. The temperature of the pond water, disturbances in water hydrochemistry, organic pollution of the ponds, as well as the physiological weakness of the fish for wintering (non-standard weight, weak fatness, etc.) cause mass diseases of young fish. In summer, the disease does not manifest itself and the chilodonellas disappear, and at this time they turn into cysts in a dormant state. The temperature of the pond water, disturbances in water hydrochemistry, organic pollution of the ponds, as well as the physiological weakness of the fish for wintering (non-standard weight, weak fatness, etc.) cause mass diseases of young fish. As the pathological process develops, a bluish-gray mucous plaque appears on the body of the fish. The gills swell, thicken, and their pattern becomes flat, and sometimes gill hyperemia is observed. The course of the disease is often complicated by other ectoparasites (trichodina, apiozomosis, costia). Often, epidemics of hildonellosis end with mass mortality of fish.

Chylodonellosis triggers: a - Chilodonella cyprinid; b-Ch. hexasticha; 1st nucleus; 2nd Cilia (Cilia) rows; 3rd oral apparatus. 1 photo.

Chilodonella cyprinid; carp fish with xylodonellosis (carp). 2 photo.

We made the diagnosis based on the results of microscopic examination of the symptoms of the disease and the smears taken from the surface of the body, fins, and gills. The presence of more than 40-50 ciliates in the field of view of the microscope indicated a severe invasion. A low number





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of parasites (5-15 or more) indicated the need for antiparasitic treatment of the fish. During the winter months, when the disease is at its peak, we treated the fish directly in ponds, cages, and other containers. We used the appropriate methods of applying the preparations. For this purpose, we used a 0.1-0.2% solution of table salt and kept our fish for 24-48 hours. To do this, we first determined the volume of water in the pool and put it in the ice hole in the same sifted volume of salt, and then also applied malachite green (solution in a ratio of 5:1000), poured it into the pool through an ice hole and processed it into the pool, yielding a concentration of 0.1-0.2 g/m³. We kept the fish in the solution for 4-5 hours. We repeated the treatment again after 1-2 days, each time controlling the effectiveness of the fish through a microscopic examination method. We continued the treatment 3-4 times. Treatment methods in pools: Salt baths (0.1-0.2%, 24-48 hours). Malachite green (0.1-0.2 g/m³, 4-5 hours), 3-4 treatments at intervals of 1-2 days.

Conclusion.

Chilodonella is a common parasitic disease in cyprinid fish, and its diagnosis, prevention, and treatment are of great importance to the fishing industry. We diagnosed the disease using microscopic examinations and other modern methods. The hydrochemistry and temperature of the pool water, as well as the use of salt and malachite green in our treatment, played a key role in improving the living conditions of the fish. By using proper prevention and treatment methods, we have reduced the spread and incidence of Chilodonella cyprinid in fish, reduced mortality, and improved fish health.

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